

## TOOLS & ENVIRONMENT

*Exploring the EDA ecosystem and setting up our  
development environment*

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# Open-Source Silicon Revolution

Traditionally, chip design has relied on expensive proprietary tools and restricted access to manufacturing technologies.

Open-source initiatives are changing this by enabling:

- ▶ Open PDKs: Access to technology information needed to design manufacturable chips.
- ▶ Open EDA tools: Community-developed tools for simulation, synthesis and physical design.
- ▶ Open hardware: Reusable processors, accelerators and hardware IP.
- ▶ Accessible fabrication: Educational programs and multi-project wafer initiatives.

# From RTL to Silicon

## 1. RTL Design

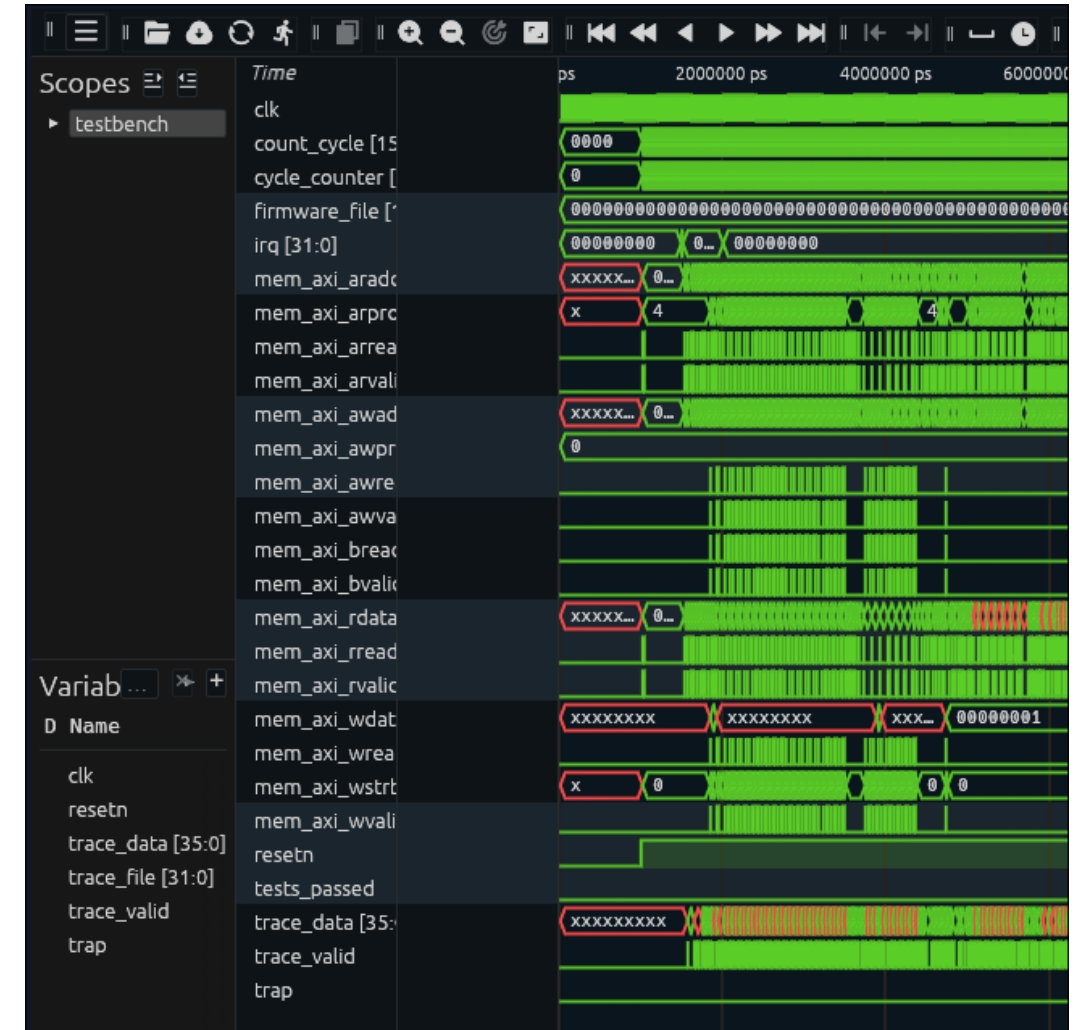
- ▶ Translate specifications and design constraints into RTL.
- ▶ Describe hardware behavior and structure using an HDL.



```
1 module full_adder (  
2     input wire a,  
3     input wire b,  
4     input wire cin,  
5     output wire sum,  
6     output wire cout  
7 );  
8  
9 assign {cout, sum} = a + b + cin;  
10  
11 endmodule
```

# From RTL to Silicon

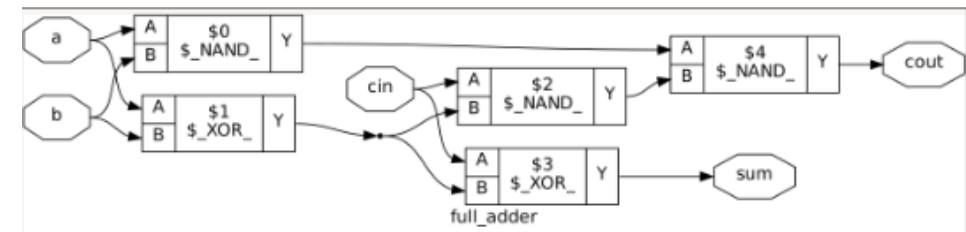
1. RTL Design
2. Functional Verification
  - ▶ Test design functionality.
  - ▶ Simulation before implementation.



# From RTL to Silicon

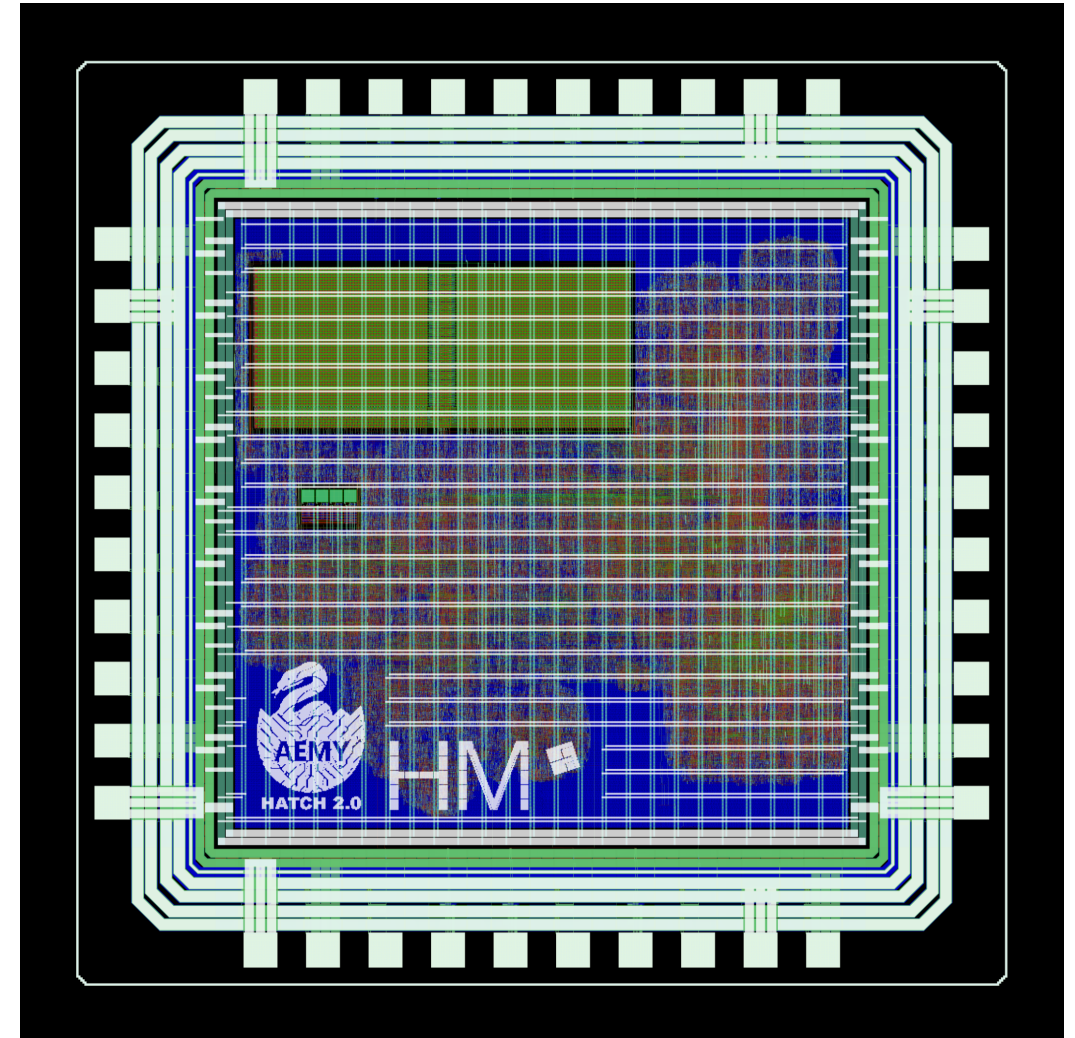
1. RTL Design
2. Functional Verification
3. Logic Synthesis
  - ▶ RTL → gate-level netlist.
  - ▶ Technology mapping.

```
1 module full_adder (  
2     input wire a,  
3     ...  
4 );  
5  
6 assign {cout, sum} = a + b + cin;  
7  
8 endmodule
```



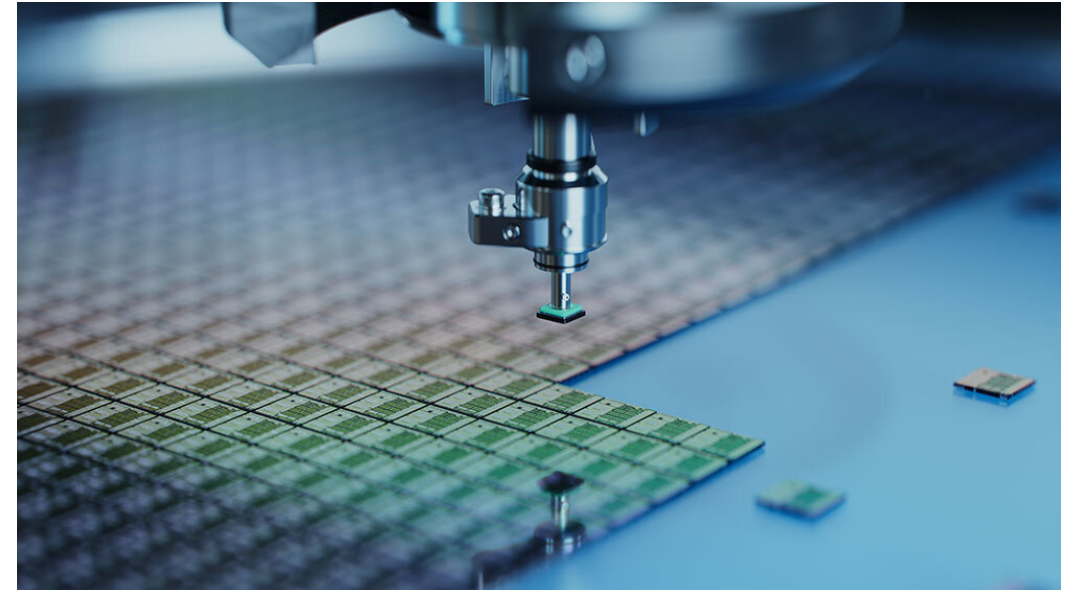
# From RTL to Silicon

1. RTL Design
2. Functional Verification
3. Logic Synhtesis
4. Physical Design
  - ▶ Placement & routing.
  - ▶ Netlist → chip layout



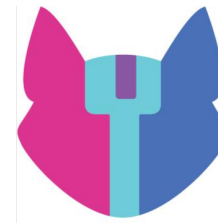
# From RTL to Silicon

1. RTL Design
2. Functional Verification
3. Logic Synthesis
4. Physical Design
5. Fabrication
  - ▶ Layout → manufacturing masks.
  - ▶ Silicon fabrication.



# The Open-Source EDA Landscape

- ▶ Each tool addresses a specific stage of the chip design flow.
- ▶ Tools can be combined into automated RTL-to-GDSII workflows.
- ▶ Open-source tools enable simulation, verification, synthesis, and physical design.
- ▶ Widely used and continuously improved by academia, industry, and the community.



YosysHQ



VERILATOR



OpenROAD

LibreLane

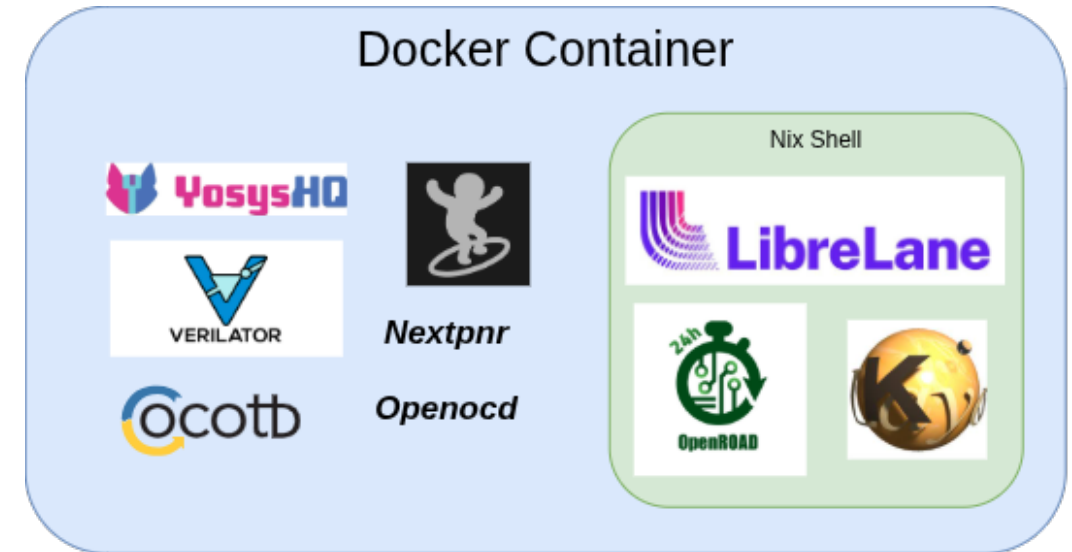
**Magic** VLSI Layout Tool  
Current distribution version 8.3





# Our Development Environment

- ▶ All students use the same preconfigured environment.
- ▶ Digital design tools are provided through OSS-CAD-Suite.
- ▶ The ASIC flow is accessed through LibreLane via Nix.
- ▶ This ensures consistency across different host operating systems.



# Getting Started

- ▶ Clone the repository using Git.
- ▶ Read **carefully** the README file.
- ▶ Follow the installation and verification instructions.



<https://gitlab.lrz.de/design-your-chip-2027/design-your-chip-environment>

## Contact

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